

National University of Computer and Emerging Sciences

Design and Analysis of Algorithms (CS2009)

Quiz - 1

Total Time: 50 Minutes

Total Marks: 2.5

Total Questions: 03

Date: October 21st 2025

Course Instructor(s)

Dr. Fahad Sherwani

Semester: FALL-2025

Campus: Karachi

Dept: School of Computing

Student Name

Roll No

Section

Student Signature

CLO 1: To understand, analyze and compute the algorithms' time complexities using the acquired knowledge.

Q1: Answer the following questions briefly.

[0.5 Mark]

- a) How to identify if a problem is well suited for dynamic programming technique to be applied?
- b) What is memoization? How is it different from tabulation?

CLO 2: To design efficient algorithms for real world scenarios.

Q2: Show a dry run of dynamic programming technique for minimizing the multiplication operations in the matrix chain given as: $P_0 = 5, P_1 = 4, P_2 = 6, P_3 = 2, P_4 = 7, P_5 = 5$. Also identify the optimal placement of brackets.

[1 Mark]

CLO 2: To design efficient algorithms for real world scenarios.

Q3: Given a knapsack of $W=5$, find the maximum value (**vi**) that can be achieved by placing the following items without exceeding the total capacity of the knapsack. Also show the keep table that identifies the items that are being selected.

[1 Mark]

Item (i)	Weight (wi)	Value (vi)
1	2	10
2	3	15
3	1	4
4	4	20

BEST OF LUCK